

Derivation of $R(2, 2)$

1 Setup

1.1 HP Residual and Transfer Functions

For closed-loop delay $d = 2$, the HP residual is

$$\delta p_{mr}(z^{-1}) = -F_T(z^{-1}) \delta x_T(z^{-1}) + F_N(z^{-1}) n_x(z^{-1}), \quad (1)$$

$$F_T(z^{-1}) = \frac{(1 - a_{pd})(1 - z^{-1}) z^{-3} [1 + (1 - \lambda_c)z^{-1} + (1 - \lambda_c)z^{-2}]}{(1 - (1 - a_{pd})z^{-1})(1 - \lambda_c z^{-1})}, \quad (2)$$

$$F_N(z^{-1}) = (1 - z^{-1}) z^3 F_T(z^{-1}), \quad (3)$$

where $\delta x_T, n_x$ are independent zero-mean Gaussian white noise with variances $\sigma_{\delta x_T}^2, \sigma_{n_x}^2$; $a_{pd} \in (0, 1)$ is the IIR LP coefficient and $\lambda_c \in (0, 1)$ is the closed-loop pole.

1.2 Variance of δp_{mr}

From the companion document,

$$\sigma_{\delta p_{mr}}^2 = C_{\text{dpmr}}(\lambda_c, a_{pd}) \sigma_{\delta x_T}^2 + C_n(\lambda_c, a_{pd}) \sigma_{n_x}^2. \quad (4)$$

1.3 Variance Estimator and Target

$$\hat{\sigma}^2[k] = (1 - a_{cov}) \hat{\sigma}^2[k-1] + a_{cov} \delta p_{mr}^2[k], \quad H_E(z^{-1}) = \frac{a_{cov}}{1 - (1 - a_{cov})z^{-1}}, \quad a_{cov} \in (0, 1). \quad (5)$$

$$R(2, 2) := \text{Var}(\hat{\sigma}^2[k]) \quad \text{at } k \rightarrow \infty. \quad (6)$$

2 Three-Layer Decomposition

$$R(2, 2) = K_{\text{var}} \cdot \sigma_{\delta p_{mr}}^4 \cdot \text{IF}, \quad (7)$$

- K_{var} : variance-estimation prefactor, function of a_{cov} only (§3).
- $\sigma_{\delta p_{mr}}^2$: from (4).
- IF : color-inflation factor (IF = 1 for white δp_{mr}), §§4–6.

3 Closed Form for K_{var}

3.1 Gaussian Fourth-Moment Factor

By Isserlis' theorem, for $X \sim \mathcal{N}(0, \sigma^2)$, $\text{Var}(X^2) = 2\sigma^4$. Define

$$g_{\text{gauss}} := \text{Var}(\delta p_{mr}^2) / \sigma_{\delta p_{mr}}^4 = 2. \quad (8)$$

3.2 Variance Gain of H_E

For a stable LTI filter H and stationary input x with PSD S_x ,

$$\text{Var}(H \cdot x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} |H(e^{j\omega})|^2 S_x(\omega) d\omega, \quad (9)$$

which for white x reduces to $\|H\|^2 \sigma_x^2$ with variance gain $\|H\|^2 := (1/2\pi) \int |H|^2 d\omega$. Applied to H_E in (5),

$$g_E := \|H_E\|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |H_E(e^{j\omega})|^2 d\omega = \frac{a_{cov}}{2 - a_{cov}}. \quad (10)$$

3.3 Composition

$$K_{\text{var}} := g_{\text{gauss}} \cdot g_E = \frac{2 a_{cov}}{2 - a_{cov}}. \quad (11)$$

4 IF as a Spectral Integral

4.1 Apply Universal Variance Formula to $\hat{\sigma}^2$

Let $v[k] := \delta p_{mr}^2[k] - \sigma_{\delta p_{mr}}^2$ be the centered squared residual, with PSD $S_v(\omega)$. Applying (9) with $H = H_E$,

$$R(2, 2) = \frac{1}{2\pi} \int_{-\pi}^{\pi} |H_E(e^{j\omega})|^2 S_v(\omega) d\omega. \quad (12)$$

Because δp_{mr} inherits the LTI poles $1 - a_{pd}$ and λ_c from F_T, F_N , v is colored (S_v non-constant).

4.2 Isserlis' Theorem at Lag τ

Define

$$R_{\delta p_{mr}}(\tau) := \text{Cov}(\delta p_{mr}[k], \delta p_{mr}[k + \tau]), \quad \rho_{\delta p_{mr}}(\tau) := R_{\delta p_{mr}}(\tau) / \sigma_{\delta p_{mr}}^2. \quad (13)$$

By Isserlis' theorem applied at lag τ for zero-mean stationary Gaussian δp_{mr} ,

$$R_v(\tau) = 2 R_{\delta p_{mr}}^2(\tau), \quad \rho_v(\tau) = \rho_{\delta p_{mr}}^2(\tau). \quad (14)$$

4.3 Result

Evaluating the integral (12) using (14) (details omitted),

$$R(2, 2) = K_{\text{var}} \cdot \sigma_{\delta p_{mr}}^4 \cdot \text{IF}, \quad \text{IF} := 1 + 2 \sum_{\tau \geq 1} \rho_{\delta p_{mr}}^2(\tau) s^\tau, \quad s := 1 - a_{cov}. \quad (15)$$

IF = 1 for white δp_{mr} ($\rho_{\delta p_{mr}}(\tau) = 0, \tau \geq 1$); the colored case is closed in §§5–6.

5 Closed Form for $\rho_{\delta p_{mr}}(\tau)$

5.1 Local Notation

$$\alpha := 1 - a_{pd}, \quad \beta := \lambda_c, \quad c := 1 - \lambda_c. \quad (16)$$

5.2 $R_{F_T}(\tau)$ via Parseval and Residues

By Parseval,

$$R_{F_T}(\tau) = \frac{1}{2\pi} \int_{-\pi}^{\pi} |F_T(e^{j\omega})|^2 \cos(\omega\tau) d\omega. \quad (17)$$

Evaluating (17) by residues at the poles $z = \alpha$ and $z = \beta$ of F_T ,

$$R_{F_T}(\tau) = A_\alpha \alpha^\tau + A_\beta \beta^\tau, \quad \tau \geq 3, \quad (18)$$

with A_α, A_β closed in (α, β, c) . Values $R_{F_T}(0), R_{F_T}(1), R_{F_T}(2)$ follow by direct evaluation of (17).

5.3 Autocorrelation $R_{F_N}(\tau)$

From (3), $|F_N(e^{j\omega})|^2 = 2(1 - \cos \omega)|F_T(e^{j\omega})|^2$. By inverse DTFT,

$$R_{F_N}(\tau) = 2 R_{F_T}(\tau) - R_{F_T}(\tau - 1) - R_{F_T}(\tau + 1). \quad (19)$$

5.4 Combined $\rho_{\delta p_{mr}}(\tau)$

By independence of $\delta x_T, n_x$,

$$R_{\delta p_{mr}}(\tau) = \sigma_{\delta x_T}^2 R_{F_T}(\tau) + \sigma_{n_x}^2 R_{F_N}(\tau), \quad \rho_{\delta p_{mr}}(\tau) = R_{\delta p_{mr}}(\tau) / \sigma_{\delta p_{mr}}^2. \quad (20)$$

For $\tau \geq 3$, both R_{F_T}, R_{F_N} are linear combinations of α^τ, β^τ , so

$$\rho_{\delta p_{mr}}(\tau) = c_\alpha \alpha^\tau + c_\beta \beta^\tau, \quad \tau \geq 3, \quad (21)$$

with c_α, c_β closed in $(\alpha, \beta, c, \sigma_{\delta x_T}^2, \sigma_{n_x}^2)$. Values $\rho_{\delta p_{mr}}(1), \rho_{\delta p_{mr}}(2)$ follow by case-split evaluation.

6 Closed Form for IF

Squaring (21),

$$\rho_{\delta p_{mr}}^2(\tau) = c_\alpha^2 (\alpha^2)^\tau + 2c_\alpha c_\beta (\alpha\beta)^\tau + c_\beta^2 (\beta^2)^\tau, \quad \tau \geq 3. \quad (22)$$

The $\tau \geq 3$ part of the sum in (15) splits into three geometric series:

$$\sum_{\tau \geq 3} (\alpha^2 s)^\tau = \frac{(\alpha^2 s)^3}{1 - \alpha^2 s}, \quad (23)$$

and likewise for $\alpha\beta s, \beta^2 s$. With the explicit $\tau = 1, 2$ terms,

$$\boxed{\text{IF} = 1 + 2 \left[\rho_{\delta p_{mr}}^2(1) s + \rho_{\delta p_{mr}}^2(2) s^2 + c_\alpha^2 \frac{(\alpha^2 s)^3}{1 - \alpha^2 s} + 2c_\alpha c_\beta \frac{(\alpha\beta s)^3}{1 - \alpha\beta s} + c_\beta^2 \frac{(\beta^2 s)^3}{1 - \beta^2 s} \right]}. \quad (24)$$

Convergence requires $\alpha^2 s, \alpha\beta s, \beta^2 s < 1$, guaranteed by $\alpha, \beta, s \in (0, 1)$.

Caveat. The form (21) is valid only for $\tau \geq 3$. Replacing the three $(\cdot)^3/(1 - \cdot)$ tails by $(\cdot)/(1 - \cdot)$ and dropping the explicit $\tau = 1, 2$ terms introduces relative errors up to $\sim 135\%$ at $a_{pd} = 0.5$.